

Laboratory 1.1: Cancer Cell Line Characterization

Good morning, everyone, and welcome to your first hands-on laboratory session in our cancer biology course. Today marks an important transition from theoretical concepts to practical application. You're going to work with real cancer cells—the same types of cells that researchers around the world use to understand cancer biology and develop new treatments.

Laboratory Overview and Safety Briefing

Before we begin, let's talk about what we're doing today and why it's important. You'll be working with three different cancer cell lines that represent major cancer types: HeLa cells from cervical cancer, MCF-7 cells from breast cancer, and HCT116 cells from colorectal cancer. These aren't just random choices—each of these cell lines has taught us fundamental lessons about cancer biology, and each represents different aspects of the hallmarks we discussed in lecture.

Now, safety first. While these are established laboratory cell lines and not infectious, we treat all biological materials with respect and caution. You'll be working in tissue culture hoods, wearing gloves at all times, and following strict sterile technique. The chemicals we're using today are generally safe, but we'll handle them properly. If you have any questions about safety at any point, please ask immediately.

Understanding Your Cell Lines: More Than Just Cancer Cells

Let me tell you a bit about each cell line you'll be working with, because understanding their backgrounds will help you interpret what you observe today.

HeLa cells are perhaps the most famous cells in biology. They were derived from Henrietta Lacks, an African American woman who died of cervical cancer in 1951. What makes HeLa cells extraordinary is their immortality—they've been growing continuously in laboratories for over 70 years. They're infected with human papillomavirus type 18, which inactivated their p53 and RB tumor suppressor genes. This gives them the ability to grow indefinitely and makes them relatively easy to culture. You'll notice they grow quite aggressively and can tolerate handling well.

MCF-7 cells come from a breast adenocarcinoma and represent a different type of cancer biology. These cells are estrogen receptor positive, which means they respond to estrogen signaling. They're less aggressive than HeLa cells and grow more slowly. They also retain some characteristics of normal breast epithelial cells, including the ability to form dome-like structures in culture. MCF-7 cells have been crucial for understanding hormone-dependent cancers.

HCT116 cells are from a colorectal carcinoma and represent yet another cancer type with distinct characteristics. These cells have mutations in key genes including KRAS and PIK3CA, making them valuable for studying oncogene-driven cancer biology. They grow at a moderate rate and have distinctive morphology that you'll observe today.

Part 1: Sterile Technique and Cell Culture Fundamentals

Let's start with the foundation of all cell culture work: sterile technique. This isn't just about preventing contamination—though that's crucial—it's about maintaining the integrity of your experimental system.

Setting Up Your Workspace

First, we'll prepare your tissue culture hood. Think of this as your sterile sanctuary. Turn on the hood and let it run for at least 10 minutes before you begin. While you wait, gather all your materials: pipettes, tips, media, flasks, and waste containers.

Now, here's the critical part: everything that goes into the hood must be sterile, and everything you do must maintain sterility. Spray your gloves and all surfaces with 70% ethanol. This concentration is important—it's more effective than higher concentrations because the water helps the ethanol penetrate microbial cell walls.

When you place items in the hood, think about workflow. Put clean items on one side, work in the middle, and have your waste on the other side. Never reach over sterile items with non-sterile materials. Your arms should move in smooth, deliberate motions—rapid movements create turbulence that can introduce contaminants.

Working with Pipettes

Pipetting in cell culture requires precision and sterility. When you remove a pipette from its wrapper, touch only the top end—never touch the tip or shaft. Insert the pipette into your pipette controller with a firm, straight motion.

When pipetting media or cell suspensions, always use a fresh pipette for each solution. Never double-dip. When you're done with a pipette, place it immediately in your waste container. This might seem wasteful, but contamination can destroy weeks of work.

Understanding Your Culture Media

The media you're using today contains everything these cancer cells need to survive and grow: amino acids, vitamins, salts, glucose, and serum. The serum—usually from fetal bovine sources—provides growth factors and proteins that cells need. The media also contains antibiotics to prevent bacterial contamination, and phenol red as a pH indicator. When the media is healthy, it should be a salmon-pink color. If it turns yellow, that indicates acidic conditions, usually from bacterial contamination or overgrown cells.

Part 2: Cell Observation and Morphological Analysis

Now let's examine your cells. Each cell line has distinct morphological characteristics that tell us about their biology.

Using the Microscope Effectively

Start with the 10x objective for an overview, then move to 20x for detailed observations. Cell culture microscopes use phase contrast, which converts small differences in refractive index into visible contrast. This allows you to see living cells clearly without staining them.

When you focus on cells, look for several key features. Healthy cells should be well-attached to the culture surface (except for the few that normally grow in suspension). They should have clear, defined boundaries and relatively uniform appearance within each cell type.

HeLa Cell Morphology

HeLa cells typically appear as irregular, somewhat elongated cells with a "fried egg" appearance when well-spread. They're relatively large and have prominent nuclei. You'll notice they don't grow in organized patterns—they tend to grow over and under each other, reflecting their aggressive, transformed nature. This growth pattern—called "loss of contact inhibition"—is one of the hallmarks of cancer cells.

Look for cells in different phases of the cell cycle. Some will be round and bright—these are likely dividing. Others will be well-spread and flattened—these are in interphase. The fact that you can see cells in all phases simultaneously tells you this is an actively growing population.

MCF-7 Cell Characteristics

MCF-7 cells look quite different. They're more epithelial in appearance, often growing in clusters or sheets. Individual cells are typically smaller than HeLa cells and more uniform in shape. If the culture is confluent, you might see them forming dome-like structures—this is a characteristic behavior that reflects their epithelial origin.

These cells maintain more contact inhibition than HeLa cells, so they're less likely to grow over each other. This reflects their somewhat less aggressive nature compared to HeLa cells.

HCT116 Cell Features

HCT116 cells have their own distinctive appearance. They're typically epithelial-like but may appear more spindle-shaped than MCF-7 cells. They often grow in patterns that reflect their colonic epithelial origin. You might notice they tend to align with each other in certain areas.

Pay attention to cell density in each culture. Different cell lines prefer different densities for optimal growth. Overcrowded cells may become stressed, while too few cells may not receive adequate paracrine signals for healthy growth.

Part 3: Cell Counting and Growth Assessment

Now we'll quantify what we're observing by counting cells and assessing their growth characteristics.

Preparing Cell Suspensions

To count cells, we first need to detach them from the culture surface and create a single-cell suspension. We'll use trypsin-EDTA, a proteolytic enzyme that cleaves the proteins holding cells to the substrate and to each other.

First, remove the media from your culture and save it—it contains any cells that were already floating. Rinse the cells gently with phosphate-buffered saline to remove residual media and serum, which can inhibit trypsin activity.

Add just enough trypsin to cover the cell monolayer—usually about 1 milliliter for a T25 flask. Too much trypsin can damage cells, while too little won't detach them effectively. Place the flask in the incubator for 2-5 minutes. Different cell lines require different incubation times—HeLa cells usually detach quickly, while MCF-7 cells may take longer.

Watch for the cells to round up and begin to detach. You can gently tap the side of the flask to help this process, but don't be too aggressive. Once most cells are floating, add an equal volume of complete media to neutralize the trypsin.

Hemocytometer Counting Technique

The hemocytometer is a precision tool that allows accurate cell counting. It consists of a thick glass slide with a grid etched into its surface. When you place a coverslip on top, it creates a chamber of known volume.

Load your cell suspension onto the hemocytometer by placing a small drop at the edge of the coverslip. Capillary action will draw the suspension under the coverslip. You want to fill the chamber completely but not overflow it.

Under the microscope, you'll see a grid pattern. Count the cells in the four corner squares of the large central grid. Count cells that are within the square or touching the top and left borders, but not those touching the bottom and right borders. This prevents double-counting at the borders.

For each cell line, count at least 100 cells total across the four squares to get reliable statistics. If you count fewer than 100 cells, your count may not be accurate due to sampling error.

Calculating Cell Concentration

The calculation seems complex at first, but it's actually straightforward. Each large square represents 1 square millimeter of area, and the depth under the coverslip is 0.1 millimeters, so each square represents 0.1 cubic millimeters or 0.0001 milliliters of volume.

Take your total count from the four squares, divide by 4 to get the average per square, then multiply by 10,000 to get cells per milliliter. Don't forget to account for any dilutions you made when preparing your counting sample.

Part 4: Viability Assessment

Counting cells tells us how many we have, but not whether they're healthy. We'll use two different methods to assess viability.

Trypan Blue Exclusion

Trypan blue is a vital dye that can't cross intact cell membranes. Living cells exclude the dye and remain clear, while dead cells take up the dye and appear blue.

Mix equal volumes of your cell suspension and 0.4% trypan blue solution. Let this sit for about 5 minutes at room temperature to allow the dye to penetrate dead cells fully.

Load this mixture onto a clean hemocytometer and count both total cells and blue (dead) cells. Calculate viability as the percentage of cells that exclude the dye. Healthy cultures should typically show greater than 90% viability.

Be aware that trypan blue only identifies cells that have lost membrane integrity—it doesn't detect cells that are alive but stressed or dying. Also, work quickly with the trypan blue mixture, as prolonged exposure can damage even healthy cells.

MTT Metabolic Assay

The MTT assay measures cellular metabolic activity as an indicator of viability and proliferation. MTT is a yellow tetrazolium salt that living cells convert to purple formazan crystals through mitochondrial enzyme activity.

Seed equal numbers of each cell line into 96-well plates—about 5,000 cells per well is usually appropriate. Use multiple wells for each cell line to account for variability. Include some wells with media only as negative controls.

After the cells have had time to attach and recover from trypsinization—usually 4-6 hours—add MTT solution to each well. The concentration should be about 0.5 mg/ml in culture media. Incubate for 2-4 hours at 37 degrees Celsius.

During this incubation, metabolically active cells will convert the yellow MTT to purple formazan crystals. The amount of purple color produced is proportional to the number of viable, metabolically active cells.

After incubation, carefully remove the media without disturbing the formazan crystals. Add dimethyl sulfoxide to dissolve the crystals, creating a purple solution whose intensity you can measure spectrophotometrically.

Part 5: DNA Extraction for Future Analysis

Finally, we'll extract DNA from a portion of each cell line for mutation analysis in upcoming labs.

Cell Harvesting for DNA Extraction

Take about 1-2 million cells from each line and pellet them by centrifugation. Remove the supernatant carefully—the cell pellet may be quite small and easy to disturb.

Wash the pellet once with phosphate-buffered saline to remove any residual media, which can interfere with DNA extraction. After centrifugation, remove as much supernatant as possible without disturbing the pellet.

DNA Extraction Process

We'll use a simple lysis method suitable for PCR-based analysis. Add lysis buffer containing detergent to break open cell membranes and protease to digest proteins. The exact composition includes Tris buffer for pH stability, EDTA to chelate metal ions that might damage DNA, and sodium chloride to help precipitate proteins.

Incubate this mixture at 55 degrees Celsius for several hours or overnight. The protease will digest cellular proteins, including histones that are tightly bound to DNA. This step is crucial for getting clean, amplifiable DNA.

After protein digestion, add cold isopropanol to precipitate the DNA. DNA is soluble in water but not in alcohol, so this step separates it from remaining cellular components. You should see white, stringy DNA precipitate forming.

Centrifuge to pellet the precipitated DNA, then wash it with 70% ethanol to remove salt residues. Finally, air-dry the pellet and resuspend it in water or TE buffer for storage.

Quality Assessment

Good DNA should be clear and viscous when resuspended. You can assess quality by running a small sample on an agarose gel—intact DNA should appear as a high molecular weight band with minimal smearing.

Store your DNA samples at -20 degrees Celsius. Properly extracted and stored DNA should remain stable for months and be suitable for PCR amplification of specific gene regions.

Data Recording and Analysis

Throughout today's session, maintain detailed records in your laboratory notebook. Record cell morphology observations, count data, viability percentages, and any unusual observations.

When comparing cell lines, consider not just the numbers but what they tell you about cancer biology. Why might HeLa cells grow faster than MCF-7 cells? What does morphology tell you about their tissue of origin? How might the mutations present in each line affect their growth characteristics?

For your laboratory report, you'll analyze these data in the context of the hallmarks of cancer we've discussed. Which hallmarks are most evident in each cell line based on your observations today?

Troubleshooting Common Issues

If your cell counts seem unusually low, consider whether trypsinization was complete. Some cells are more adherent than others and may need longer incubation or gentle agitation.

If viability is poor, think about potential causes: old media, contamination, improper incubator conditions, or rough handling during processing.

If cells look unhealthy under the microscope—granular, detached, or irregular—consider environmental factors like pH, temperature, or CO₂ levels.

Connecting to Course Concepts

As you work today, think about how these observations connect to our theoretical framework. The unlimited growth potential you're observing represents the "enabling replicative immortality" hallmark. The loss of contact inhibition in HeLa cells reflects "evading growth suppressors." The fact that these cells can grow in artificial media shows their independence from normal growth signals.

Each cell line has acquired these capabilities through different molecular mechanisms, which we'll explore in detail in upcoming lectures and labs. The DNA you're extracting today will allow us to examine some of these mechanisms directly.

Preparation for Next Week

Clean and organize your workspace thoroughly. Contamination prevention is everyone's responsibility. Store your DNA samples properly labeled and documented.

Review your observations and data before our next session. We'll be using your cell counting data to set up experiments examining oncogene and tumor suppressor status in these same cell lines.

Think about questions that arose during today's work—we'll discuss these at the beginning of our next laboratory session.

Remember, cell culture is both an art and a science. The techniques you're learning today form the foundation for virtually all cancer research. Master these basics, and you'll be prepared for increasingly sophisticated experiments as we progress through the course.

Good luck, work carefully, and don't hesitate to ask questions. Your success today sets the stage for everything we'll do in the laboratory component of this course.